

Pareto-Improving Pricing: Why 3 Is Better Than 2^{*}

Zi Yang Kang[†]  Piotr Dworczak[‡]

September 2026

Abstract

We study the design of priority pricing systems with heterogeneous agents in environments in which improving quality for some agents reduces the average quality that can be provided. Contrary to the equity–efficiency tradeoff emphasized in public debates, we show that under economically natural conditions priority pricing can Pareto-improve on an equal-allocation benchmark. Three priority tiers suffice for such an improvement, combining higher quality for a fee, lower quality with compensation, and an intermediate tier at the benchmark quality; two tiers are never enough. Our results provide a framework for overcoming equity–efficiency tensions in applications such as lane pricing, waiting-line design, public provision, and insurance.

JEL classification: D82, D47, D63

Keywords: Pareto improvements, mechanism design, inequality-aware market design

^{*} We thank Simon Loertscher, Axel Ockenfels, Paula Onuchic, Mike Ostrovsky, Agathe Pernoud, Bruno Strulovici, Filip Tokarski, Shosh Vasserman, Frank Yang, and several seminar audiences for helpful discussions. We gratefully acknowledge the support received under the ERC Starting Grant IMD-101040122. This research was conducted in part at the Adenauer School of Government (ASG) at the University of Cologne with generous support from the Alfred Landecker Foundation.

[†] Department of Economics, University of Toronto; zy.kang@utoronto.ca.

[‡] University of Cologne; Northwestern University; and Group for Research in Applied Economics (GRAPE); dworczak.piotr@gmail.com.

1 Introduction

Many scarce resources—such as space on highways, bandwidth in communication networks, and access to public services—are allocated through waiting rather than explicit prices. Since at least [Pigou \(1920\)](#), economists have emphasized the inefficiency of such time-based allocation: waiting is socially costly, and many individuals would pay to avoid it, creating scope for gains from trade. Yet pricing is often resisted on equity grounds because it is seen as replacing time-based rationing with income-based rationing.

Public debates surrounding “pay-to-skip-the-line” systems vividly reflect this concern. On roads, critics deride priority toll lanes as “Lexus lanes” that allow the rich to buy their way out of congestion while others bear the cost of greater delays ([Anderson, 2005](#)). At airports, paid access to expedited security lines—such as CLEAR—has been portrayed as allowing some passengers to cut in line ([Mull, 2023](#)) and as splitting “travelers into haves and have-nots” ([Stewart, 2022](#)). Similar concerns arise when governments offer fast-track treatment for payment. Critics of investor-citizenship programs, for example, describe them as allowing wealth to purchase accelerated access to a public status for which ordinary applicants must wait ([Shachar, 2018](#)).

Our main insight in this paper is that this equity–efficiency tradeoff need not be as stark as these debates suggest. We define the *equity benchmark* as an allocation in which all individuals receive the same quality (e.g., face the same wait time). We assume that departures from this benchmark are costly: giving some individuals higher quality requires both giving others lower quality and lowering quality on average. Even so, prices and priority can be introduced in a way that *Pareto-improves* on the equity benchmark, but only by moving beyond the two-tier systems that dominate practice. With only two tiers—a paid “fast lane” and a free regular option—no Pareto improvement is possible: even if the revenue from selling priority is redistributed, some individuals must be worse off than under the equity benchmark. With three tiers, by contrast, every individual can be made strictly better off. Our construction combines a paid high-priority tier, a compensated low-priority tier, and an intermediate tier that preserves the benchmark allocation. Together, these results help explain why simple priority systems generate backlash: without a third tier, unequal priority necessarily harms some individuals. At the same time, our framework identifies designs under which everyone benefits from the introduction of prices and priority.

Our formal model is stylized but flexible enough to cover a wide range of applications. We consider a designer who allocates potentially different qualities of a good to a unit mass of agents with different marginal rates of substitution between money and quality. In lane pricing, for example, quality decreases with travel time, and agents vary in their distaste for waiting. Under the equity benchmark, every agent receives the same quality and makes no monetary transfer. The designer seeks to replace this benchmark with a pricing mechanism—a menu of qualities and monetary transfers—that introduces dispersion in quality. Application-specific feasibility constraints determine which quality distributions can be implemented. Rather than model these constraints directly, we impose three assumptions on the feasible set of quality distributions.

First, we assume there is a tradeoff—what we call *friction*—between dispersion and average quality. Specifically, the degenerate quality distribution associated with the equity benchmark uniquely maximizes average quality. This assumption captures the idea that reallocating quality to create dispersion is costly. For example, if travel time is convex in lane traffic, unequal lane loads create dispersion in travel times while raising average travel time. In other applications, friction may arise from curvature in preferences, or from heterogeneity in the costs of serving different types, as under adverse selection.

Second, we assume that this mean–dispersion tradeoff is locally mild. More precisely, the loss in average quality from creating dispersion can be made arbitrarily small relative to the aggregate quality gain received by agents allocated qualities above the equity benchmark. Small departures from equal lane loads, for instance, generate first-order dispersion but only a higher-order increase in average travel time.

Third, we assume that the designer can randomize assignments. Formally, if a quality distribution is feasible, then every mean-preserving contraction of that distribution is also feasible.¹

Under these assumptions, we prove two main results. On the one hand, no mechanism with only two tiers—that is, two quality levels—can Pareto-improve on the equity benchmark. On the other hand, there exists a three-tier mechanism that strictly Pareto-improves on the equity benchmark. Thus, three tiers are the minimum required to make everyone better off—a modest but practically feasible level of complexity.

For intuition, note that priority pricing creates *sorting gains* by reallocating quality toward agents with higher willingness to pay, but the resulting dispersion lowers average quality. A Pareto improvement thus

¹ In our applications, we also establish when the Pareto improvement can be implemented without randomizing assignments.

requires extracting enough of these gains through prices and redistributing the revenue to compensate every agent for the loss in average quality.

With only two tiers, revenue from the high tier finances a subsidy to the low tier. The threshold type is indifferent between the two tiers, so the price premium for the higher tier must equal this type's willingness to pay for the quality upgrade. Budget balance then implies that the subsidy exactly compensates the threshold type for receiving the lower quality rather than the mechanism's *average* quality. Consequently, the threshold type obtains precisely the value of average quality. Since average quality is lower than under the equity benchmark, the threshold type is strictly worse off.

Adding a third tier relaxes this tension by creating two distinct pricing margins. The two quality upgrades are priced at the willingness to pay of distinct threshold types, so quality improvements at the top are valued at a higher rate than the quality reductions at the bottom. This creates additional sorting gains that can be extracted and redistributed. As long as the threshold types remain bounded away from each other, these gains are first order in the size of the quality dispersion, while the corresponding loss in average quality is of higher order. The gains can therefore compensate every agent for the loss in average quality.

Our applications illustrate the breadth and flexibility of the framework. Our assumptions, and hence the main results, apply across environments with different allocation technologies and sources of friction. Beyond our running example of lane pricing, we study the design of wait lines—for instance, for airport screening or access to public services—where our results explain how a design with three differentiated lines can avoid the equity–efficiency tradeoff, whereas a two-tier system cannot. In the problem of allocating goods with heterogeneous quality, such as public housing, we show that offering at least three types of units—for example, differentiated by size—can generate a Pareto improvement over uniform provision. Finally, we consider insurance design under adverse selection and show how our framework and main result extend to environments in which the resource cost of quality depends on the recipient.

The remainder of this paper is organized as follows. We conclude this section by reviewing the related literature. Section 2 introduces the formal model, and Section 3 presents the main results. We apply and generalize these results in Section 4, and Section 5 concludes.

1.1 Related Literature

The pricing of priority in access to goods and services—and its distributional consequences—has received sustained attention in economics. As we discuss below, several papers show that priority pricing can Pareto-improve on equal access or laissez-faire in particular environments.

Our contribution to this literature is to propose a general framework for priority pricing and to characterize the minimum number of tiers required for a Pareto improvement. We model the supply side through the set of feasible quality distributions, which isolates two channels through which priority pricing affects welfare: (i) changes in average quality and (ii) sorting gains from allocating quality to agents with higher willingness to pay. We focus on environments with a mean–dispersion tradeoff, in which creating the dispersion needed for sorting *lowers* average quality, so any Pareto improvement must be driven by sorting gains. To the best of our knowledge, our paper is the first to show that three, rather than two, priority tiers are necessary to achieve a Pareto improvement in such environments.

[Chao and Wilson \(1987\)](#) and [Gershkov and Schweinzer \(2010\)](#) show that selling priority and redistributing the resulting revenue can generate Pareto improvements. [Gershkov and Schweinzer](#) study a model in which agents differ in marginal utility of time and hold property rights to potentially random positions in a queue. They characterize when these positions can be efficiently reordered through an incentive-compatible, budget-balanced mechanism while leaving every agent weakly better off in expectation. Most relevant for us, they show that a random queue order—which corresponds to our equity benchmark—can be Pareto-improved upon.² With total waiting time fixed and linear disutility from waiting, their queue-reordering model permits sorting gains without reducing average quality. Thus, it does not feature the mean–dispersion tradeoff that is at the heart of our results.

A sizable literature studies the distributional consequences of congestion pricing in the bottleneck model of [Vickrey \(1969\)](#), including [Arnott, De Palma and Lindsey \(1994\)](#), [van den Berg and Verhoef \(2011\)](#), [Hall \(2018\)](#), and [Bobbio, Cramton and Ockenfels \(2021\)](#). These models likewise do not generally feature the mean–dispersion tradeoff captured by our friction assumption. In fact, because agents can adjust their departure times in response to prices in the [Vickrey](#) framework, congestion pricing can

² [Gershkov and Winter \(2023\)](#) make the complementary point that priority service can uniformly harm consumers when access is controlled by a profit-maximizing monopolist.

reduce average congestion—or, in the language of our paper, *increase* average quality. This channel is often central to obtaining a Pareto improvement and explains why some of these papers obtain Pareto improvements even before redistributing revenue. [Hall \(2018\)](#), for example, shows that preventing the fall in road throughput associated with hypercongestion can yield a Pareto improvement even with identical agents and before revenue is redistributed. Our friction assumption, by contrast, requires dispersion to reduce average quality relative to the equity benchmark. Pareto improvements must therefore be driven by sorting gains, with pricing revenue redistributed so as to compensate every agent for the resulting loss in average quality.³

More broadly, our paper contributes to a growing market-design literature on congestion pricing, including work by [Ostrovsky and Schwarz \(2018\)](#), [Cramton, Geddes and Ockenfels \(2019\)](#), [Beheshtian, Geddes, Rouhani, Kockelman, Ockenfels, Cramton and Do \(2020\)](#), and [Ostrovsky and Yang \(2024\)](#).⁴

Finally, our paper relates to the recent literature on *inequality-aware market design*. Building on classical work by [Weitzman \(1977\)](#), [Spence \(1977\)](#), and [Nichols and Zeckhauser \(1982\)](#), this literature studies mechanism design under redistributive objectives; contributions include work by [Condorelli \(2013\)](#), [Dworczak & Kominers & Akbarpour \(2021\)](#), [Kang \(2023\)](#), and [Akbarpour & Dworczak & Kominers \(2024\)](#), among many others. The paper most closely related to ours within this literature is [Kang and Watt \(2026\)](#). They study public provision when agents also have access to a private market. The private option imposes a constraint mathematically similar to Pareto improvement. While this literature characterizes optimal mechanisms for a specified social welfare function, we instead identify conditions under which a Pareto improvement exists without specifying interpersonal welfare comparisons.

³ Because our positive result predicts a *strict* Pareto improvement, it remains valid when transfers are imperfect and a sufficiently small share of revenue is dissipated.

⁴ An active empirical literature also quantifies the welfare effects of congestion pricing; see, for example, [Hall \(2021\)](#), [Kreindler \(2024\)](#), [Cook and Li \(2025\)](#), [Cook, Kreidieh, Vasserman, Allcott, Arora, Tomkins, Turkel and van Sambeek \(2026\)](#), [Ater, Ross, Shany and Vasserman \(2026\)](#).

2 Framework

We develop an abstract framework for the design of Pareto-improving pricing systems. The framework is intentionally parsimonious: it is broad enough to encompass a wide range of applications, yet sufficiently structured to isolate the key economic forces that drive our results.

2.1 Setup

There is a unit mass of agents, each of whom demands one unit of a good. Units of the good may differ in their physical quality $q \in \mathbf{R}_+$. Each agent privately observes his type r . The distribution of types in the population is G , which has positive continuous density g on a compact interval $[\underline{r}, \bar{r}] \subset \mathbf{R}_+$. An agent of type r who is assigned a unit of physical quality q and makes a payment $p \in \mathbf{R}$ obtains utility

$$rv(q) - p,$$

where a negative payment is interpreted as a rebate to the agent. The function $v : \mathbf{R}_+ \rightarrow [0, 1]$ is assumed to be continuously differentiable, increasing,⁵ and concave, with $v(0) = 0$.

We refer to $Q := v(q)$ as the good's *effective quality* (or simply *quality*, when there is no ambiguity). Agent utility is $rQ - p$, so the agent's type is his marginal rate of substitution between effective quality and money. The distinction between physical and effective quality separates the objective characteristics of the good from their utility consequences for agents. In particular, agents are risk-neutral over lotteries in effective quality, but may be risk-averse over lotteries in physical quality (when v is strictly concave).

We use the following running example throughout the paper to illustrate the abstract framework.

Example. *Agents are commuters traveling from a common origin to a common destination. A trip entails time $t \in [\underline{t}, \bar{t}]$ spent in traffic, where $0 \leq \underline{t} < \bar{t} < 1$. Normalizing the physical quality of a trip with no time spent in traffic to 1, define the physical quality of a trip with travel time t as $q = 1 - t$. Thus, shorter travel time corresponds to higher physical quality. When v is strictly concave, agents are risk-averse over travel time: they strictly prefer a certain travel time to any nondegenerate lottery with the same mean.*

⁵ Throughout, we use “increasing” and “decreasing” to mean strictly increasing and strictly decreasing, respectively; weak monotonicity is denoted by “nondecreasing” and “nonincreasing.”

2.2 Outcomes

A designer assigns each agent a possibly degenerate lottery over physical qualities. Because utility is linear in effective quality Q , an agent's payoff depends on his assigned lottery only through its expected quality. We can therefore summarize the physical allocation by a distribution $F \in \Delta([0, 1])$ of expected qualities available for assignment, where $1 - F(Q)$ is the mass of agents who can be assigned expected quality strictly greater than Q .⁶

Different applications restrict the feasible distributions of expected qualities in different ways. We represent these restrictions by a set $\mathcal{F} \subseteq \Delta([0, 1])$ from which the designer must select F .⁷ The set $\mathcal{F}$ is the central object of our framework; Section 2.4 introduces assumptions on $\mathcal{F}$ that generate the tradeoff between dispersion and average quality underlying our results.

Each feasible distribution $F \in \mathcal{F}$ uniquely determines, up to a G -null set, a budget-balanced and incentive-compatible deterministic reduced-form mechanism in expected qualities and payments.⁸ Incentive compatibility requires expected quality to be nondecreasing in type (Myerson, 1981). Hence, the allocation rule is

$$Q(r) = F^{-1}(G(r)),$$

the unique nondecreasing allocation rule whose distribution is F , up to G -null sets. Given this allocation rule, the envelope formula determines payments up to an additive constant (Milgrom and Segal, 2002):

$$p(r) = rQ(r) - U - \int_{\underline{r}}^r Q(s) \, ds,$$

where U is uniquely chosen to satisfy budget balance: $\int_{\underline{r}}^{\bar{r}} p(r) \, dG(r) = 0$. We say that the resulting mechanism $(Q(\cdot), p(\cdot))$ implements F .

⁶ This representation remains without loss of generality when goods are scarce: an agent who receives no good can be treated as receiving physical quality zero.

⁷ For simplicity, we abstract from variable production costs. Section 4.4 shows how our results accommodate such costs.

⁸ Here, “deterministic” refers to the reduced-form pair (Q, p) assigned to each type and does not rule out randomization over physical qualities, which can be incorporated into $\mathcal{F}$. This restriction is without loss because utility is linear in expected quality and payments.

2.3 Pareto Improvements

We define the *equity benchmark* as the degenerate quality distribution δ_{Q_0} , where

$$Q_0 := \sup_{F \in \mathcal{F}} \mathbf{E}_F[Q].$$

We assume that $\delta_{Q_0} \in \mathcal{F}$.⁹ Under the equity benchmark, average quality is maximized while every agent receives expected quality Q_0 and pays zero; the resulting direct mechanism is budget-balanced and incentive-compatible. As we show in Section 4, this benchmark often coincides with the laissez-faire non-price allocation—that is, the equilibrium allocation in a system without monetary transfers.

The designer seeks to construct a Pareto improvement over the equity benchmark. For any distribution $F \in \mathcal{F}$, we say that F is a *Pareto improvement over the equity benchmark* (or simply a *Pareto improvement*) if its implementing mechanism $(Q(\cdot), p(\cdot))$ satisfies

$$rQ(r) - p(r) \geq rQ_0, \quad \text{for every } r \in [\underline{r}, \bar{r}], \quad (\text{PI})$$

with strict inequality for a positive mass of agents. We say that F is a *strict Pareto improvement* if every agent strictly prefers its implementing mechanism to the equity benchmark—that is, if the inequality in (PI) is strict for every type $r \in [\underline{r}, \bar{r}]$.

Our focus on Pareto improvements reflects an intentionally conservative welfare criterion. We make no interpersonal comparisons of utility and posit no social welfare function. Unlike much of the existing literature, which studies optimal allocations under explicit welfare objectives, we ask only whether prices can realize Pareto gains, subject to budget balance and incentive compatibility, without invoking distributional judgments.

2.4 Feasible Policies

We complete the description of our framework by imposing three economically motivated restrictions on the set of feasible expected-quality distributions $\mathcal{F} \subseteq \Delta([0, 1])$.

⁹ This assumption is without loss of generality if $\mathcal{F}$ is compact and the designer can randomize.

We first motivate these restrictions using our running example.

Example. *There are $N \geq 2$ lanes on a highway. If a mass m_i of agents is assigned to lane i , each agent in that lane experiences travel time $w(m_i)$, where $w : [0, 1] \rightarrow [\underline{t}, \bar{t}]$ is continuously differentiable, increasing, and convex. These assumptions capture congestion effects: travel time increases with lane usage at a nondecreasing rate.*

Given a vector of lane loads $\mathbf{m} = (m_1, \dots, m_N) \in \Delta^{N-1}$, the distribution of travel times is $\sum_{i=1}^N m_i \delta_{w(m_i)}$. Because travel time t corresponds to effective quality $v(1 - t)$, the induced distribution of effective qualities is

$$F^{\mathbf{m}} := \sum_{i=1}^N m_i \delta_{v(1-w(m_i))}.$$

Starting from any lane-load vector $\mathbf{m}$, the designer can randomize agents' assignments across lanes while preserving the marginal distribution of realized physical qualities. The distributions of expected qualities that can be generated this way are precisely the mean-preserving contractions of $F^{\mathbf{m}}$, which we denote by $\text{MPC}(F^{\mathbf{m}})$.¹⁰ Consequently, the feasible set of expected quality distributions is

$$\mathcal{F} = \{F \in \Delta([0, 1]) : F \in \text{MPC}(F^{\mathbf{m}}) \text{ for some } \mathbf{m} \in \Delta^{N-1}\}.$$

Rather than committing to a particular technology, we impose three axioms on the feasible set $\mathcal{F}$.

Axiom 1 (Randomization). *The feasible set $\mathcal{F}$ is closed under mean-preserving contractions: if $F \in \mathcal{F}$, then $\text{MPC}(F) \subseteq \mathcal{F}$.*

Axiom 1 formalizes the designer's ability to randomize assignments. Starting from any feasible distribution, the designer can reduce the dispersion in agents' expected qualities without changing their average. The axiom holds by construction in the running example.

¹⁰ Strassen's theorem (Müller and Stoyan, 2002, Theorem 3.4.2) implies that there exist random variables X and $\bar{X}$ such that $X \sim F$, $\bar{X} \sim \bar{F}$, and $\mathbf{E}[X | \bar{X}] = \bar{X}$ if and only if $\bar{F}$ is a mean-preserving contraction of F . Here, X is realized effective quality and $\bar{X}$ is expected effective quality. For any $F \in \Delta([0, 1])$, define

$$\text{MPC}(F) := \left\{ \bar{F} \in \Delta([0, 1]) : \int_0^x \bar{F}^{-1}(u) du \geq \int_0^x F^{-1}(u) du \text{ for every } x \in [0, 1], \text{ with equality at } x = 1 \right\}.$$

Axiom 2 (Friction). *The equity benchmark δ_{Q_0} uniquely maximizes average quality: for every $F \in \mathcal{F}$, if $F \neq \delta_{Q_0}$, then $\mathbf{E}_F[Q] < Q_0$.*

Axiom 2 captures a friction in the allocation technology: any feasible departure from the equity benchmark lowers average quality. The axiom therefore imposes a local tradeoff between dispersion and average quality around the benchmark, thereby making improvements over the equity benchmark more difficult to achieve.

The axiom holds in the running example under the maintained assumptions. To see this, define the function $\phi(m) := mv(1 - w(m))$, and note that the average quality generated by a lane-load vector $\mathbf{m}$ is $\sum_{i=1}^N \phi(m_i)$. Since w is increasing and convex and v is increasing and concave, ϕ is strictly concave. By Jensen's inequality, for any $\mathbf{m} \in \Delta^{N-1}$,

$$\sum_{i=1}^N \phi(m_i) \leq N\phi\left(\frac{1}{N}\right) = v\left(1 - w\left(\frac{1}{N}\right)\right),$$

with equality if and only if $m_i = 1/N$ for every $i \in \{1, \dots, N\}$. Hence, $Q_0 = v(1 - w(1/N))$ satisfies the requirements of the axiom, and the equal assignment of agents to lanes induces the unique feasible distribution with average quality Q_0 . Intuitively, because travel times increase with lane usage at a nondecreasing rate and agents are weakly risk-averse over travel time, equalizing lane usage uniquely maximizes average effective quality.¹¹

Axiom 3 (Smoothness). *The tradeoff between dispersion and average quality around the equity benchmark δ_{Q_0} is locally mild: for every $\varepsilon > 0$, there exists $F \in \mathcal{F}$ such that*

$$\mathbf{E}_F[(Q - Q_0)_+] > 0 \quad \text{and} \quad \frac{Q_0 - \mathbf{E}_F[Q]}{\mathbf{E}_F[(Q - Q_0)_+]} \leq \varepsilon. \quad (\text{S})$$

Axiom 3 provides a counterpoint to Axiom 2. Whereas friction requires every feasible departure from the equity benchmark to lower average quality, smoothness requires that this loss, $Q_0 - \mathbf{E}_F[Q]$, can be made arbitrarily small relative to the cumulative improvement $\mathbf{E}_F[(Q - Q_0)_+]$ above Q_0 . If the

¹¹ Note that neither strict risk aversion nor strictly convex congestion effects are required for Axiom 2 to hold.

uniqueness requirement in Axiom 2 were violated, so that a nondegenerate feasible distribution had average quality Q_0 , condition (S) would hold trivially.

Axiom 3 also holds in the running example. Starting from equal lane loads, transfer a mass η of agents from one lane to another. The first-order effects on average quality cancel because w and v are continuously differentiable. The resulting loss in average quality is therefore $o(\eta)$, whereas the cumulative improvement over Q_0 is of order η because effective quality is locally strictly decreasing in lane usage. Hence, the ratio in condition (S) converges to zero as $\eta \downarrow 0$. Section 4.1 and Appendix A.1 provide a formal proof.

3 Main Results

Our main results characterize the minimum number of tiers required to construct a Pareto improvement. To formalize simplicity in a policy-relevant way, we count the number of distinct expected-quality levels offered by the mechanism.

Definition 1. A distribution $F \in \mathcal{F}$ offers n tiers if $|\text{supp}(F)| = n$.

Any Pareto improvement must offer at least two tiers: a one-tier mechanism assigns the same quality $Q \leq Q_0$ to every agent, while incentive compatibility and budget balance require payments to be zero.

3.1 Statement and Discussion of Main Results

Our first result shows that two tiers do not suffice.

Theorem 1. Under Axiom 2, there does not exist a Pareto improvement with two tiers.

Our second result shows that three tiers suffice.

Theorem 2. Under Axioms 1 and 3, there exists a strict Pareto improvement with three tiers.

Together, Theorems 1 and 2 establish that, under our three axioms, three is the minimum number of tiers required for a Pareto improvement.

These results are particularly relevant because two-tier priority systems are widespread in practice. Examples include express lanes on roads and fast-track options in airport security, visa processing, and health care. Such systems are appealing for their simplicity and are often defended as a way to generate allocative gains by allowing agents who value priority more highly to pay for it.

In this light, Theorem 1 is a negative result: any nontrivial two-tier priority system must make some agents strictly worse off than under the equity benchmark. Relative to giving every agent the mechanism's average quality at no charge, sorting by willingness to pay benefits everyone except the agent indifferent between the two tiers. This agent receives no sorting gain to offset the loss in average quality and is therefore worse off. Even without Axiom 2, a two-tier system cannot make every agent *strictly* better off.

The negative result of Theorem 1 therefore helps explain why many two-tier priority systems have generated public backlash. When congestion or other frictions create a tradeoff between dispersion and average quality, allowing some users to purchase priority necessarily harms others, even if the resulting revenue is redistributed. Moreover, if type measures the marginal value of reduced waiting time relative to the marginal value of money (e.g., Dworczak et al., 2021), the intermediate types harmed by a two-tier system may include low- and middle-income users who place a high value on time but also have a high marginal value of money.

Theorem 2 provides the positive counterpart: three tiers are sufficient to make every agent *strictly* better off. We construct a mechanism with an intermediate tier that provides expected quality exactly equal to Q_0 . This tier separates the threshold between the low and intermediate tiers from the threshold between the intermediate and high tiers. The quality jump at each threshold generates sorting gains for the agent indifferent at the other, so that together they generate sorting gains for every agent. Axioms 1 and 3 allow us to construct such a mechanism with gains that outweigh the loss in average quality.

This positive result shows that achieving a Pareto improvement does not require a complex or finely targeted pricing system: under our three axioms, three appropriately designed priority tiers suffice. Policy debates framed as a binary choice between uniform service and a two-tier priority system are therefore unnecessarily restrictive.

Example. *Because our running example satisfies Axioms 1 to 3, Theorems 1 and 2 apply. A conventional two-tier lane-pricing system cannot Pareto-improve on equal access: creating one faster lane necessarily*

leaves some commuters strictly worse off, even if toll revenues are redistributed. By contrast, three tiers of expected quality can make every commuter strictly better off. Section 4.1 returns to this application and studies its implementation in more detail.

3.2 Proofs of Main Results

We begin by decomposing each agent's utility gain from priority pricing into two components: the effect of the change in average quality and the gains from sorting. For any feasible distribution F and its implementing mechanism (Q, p) , incentive compatibility and budget balance imply that

$$\begin{aligned}
 rQ(r) - p(r) - rQ_0 = & \underbrace{r [\mathbf{E}_F[Q] - Q_0]}_{\text{effect of change in average quality}} \\
 & + \underbrace{\int_{\underline{r}}^r (r - s) G(s) dQ(s) + \int_r^{\bar{r}} (s - r) [1 - G(s)] dQ(s)}_{\text{gains from sorting}}. \quad (1)
 \end{aligned}$$

Here, dQ denotes the Stieltjes measure induced by the nondecreasing allocation rule Q .

The decomposition shows that each interior quality jump generates sorting gains for every type except the threshold type itself. The two integrals in equation (1) measure these gains relative to giving all agents the mechanism's average quality at no charge. Jumps at thresholds below r contribute positively to the first integral, and jumps at thresholds above r contribute positively to the second.

With only two tiers, the single quality jump leaves its threshold type with no sorting gain. In Figure 1, this is the type r_0 at which the two linear segments of the utility schedule meet. By equation (1), his utility is therefore $r_0 \mathbf{E}_F[Q]$. Axiom 2 implies that $\mathbf{E}_F[Q] < Q_0$, so this point lies strictly below the utility schedule under the equity benchmark.

Proof of Theorem 1. We begin by proving equation (1) for any feasible distribution F and its implementing mechanism (Q, p) . By the envelope theorem (Milgrom and Segal, 2002), incentive

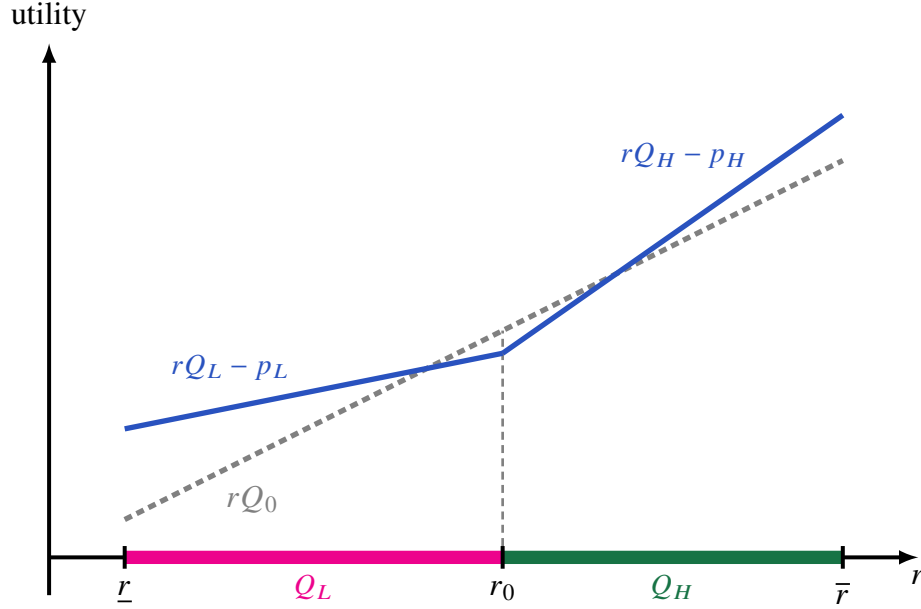


Figure 1: Agent utility in a two-tier priority system. The type r_0 is strictly worse off than under the equity benchmark.

compatibility implies that, for every $r, s \in [\underline{r}, \bar{r}]$,

$$rQ(r) - p(r) = sQ(s) - p(s) + \int_s^r Q(t) dt.$$

Averaging over s and using budget balance gives

$$\begin{aligned} rQ(r) - p(r) &= \int_{\underline{r}}^{\bar{r}} sQ(s) dG(s) + \int_{\underline{r}}^{\bar{r}} \left[\int_s^r Q(t) dt \right] dG(s) \\ &= \int_{\underline{r}}^{\bar{r}} sQ(s) dG(s) + \int_{\underline{r}}^r G(s)Q(s) ds - \int_r^{\bar{r}} [1 - G(s)]Q(s) ds. \end{aligned}$$

Stieltjes integration by parts then yields the two sorting terms in equation (1).

Every feasible two-tier mechanism has an interior threshold $r_0 > 0$, so our argument above using equation (1) applies. By continuity of the utility schedule, a positive mass of agents near r_0 is strictly worse off than under the equity benchmark. $\square$

Even without Axiom 2, the threshold type cannot be strictly better off, since $\mathbf{E}_F[Q] \leq Q_0$ by the definition of Q_0 .

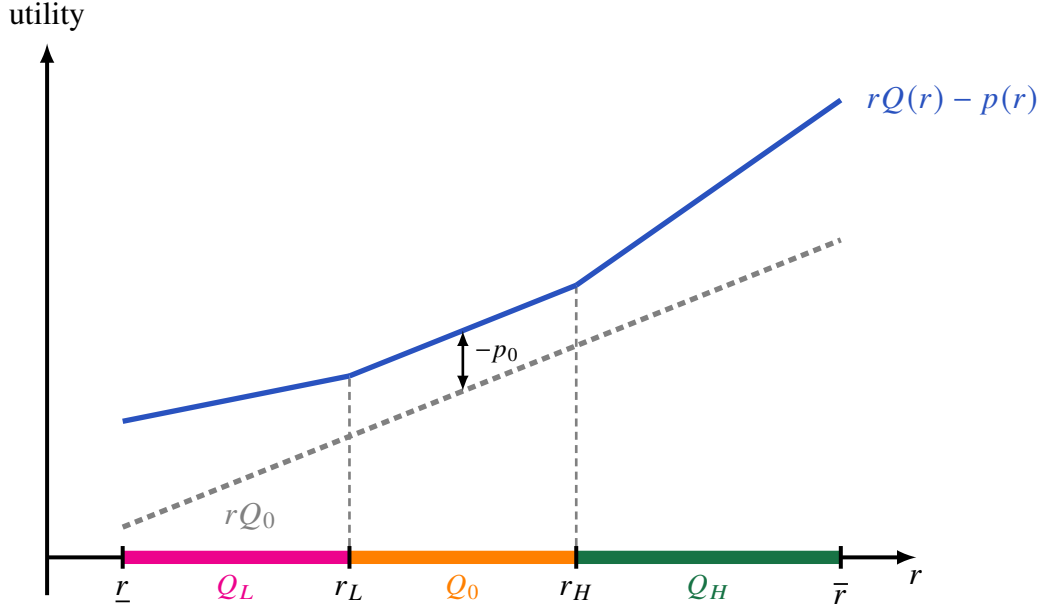


Figure 2: Agent utility in a three-tier Pareto improvement.

Corollary 1. *There does not exist a strict Pareto improvement with two tiers.*

For a three-tier mechanism whose intermediate quality is Q_0 , a strict Pareto improvement occurs precisely when the intermediate tier is subsidized. Consider a feasible distribution F with qualities $Q_L < Q_0 < Q_H$, as illustrated in Figure 2. Let $r_L < r_H$ denote its threshold types and p_0 the payment in the intermediate tier. Utility gains relative to the equity benchmark decrease with type below r_L , remain constant between r_L and r_H , and increase above r_H , so they are smallest for agents in the intermediate tier. Their utility gain is $-p_0$, the vertical difference between the two utility schedules in Figure 2. Writing $\pi_H = 1 - G(r_H)$ for the mass assigned to the high tier, equation (1) evaluated at r_L gives

$$-p_0 = (r_H - r_L) \pi_H (Q_H - Q_0) - r_L [Q_0 - \mathbf{E}_F[Q]]. \quad (2)$$

The separation between the threshold types allows sorting gains to offset the loss in average quality. At r_L , the jump to the intermediate tier contributes no sorting gain, while the jump to the high tier contributes the first term in equation (2). Because $r_H > r_L$, the quality gain in the high tier can be priced at a higher marginal willingness to pay than is required to compensate for the quality loss in the low tier.

Our construction keeps the threshold types separated while making the loss in average quality small relative to the sorting gains. We first use Axiom 3 to select a feasible distribution with a small loss in average quality relative to the cumulative improvement above Q_0 . We then use Axiom 1 to contract it into three tiers, with intermediate quality Q_0 . The contraction preserves average quality and retains enough of the cumulative improvement to finance a subsidy for the intermediate tier.

Proof of Theorem 2. Fix the threshold types $r_L := G^{-1}(1/3)$ and $r_H := G^{-1}(2/3)$. Choose $\varepsilon > 0$ sufficiently small, such that

$$0 < \varepsilon < \frac{r_H - r_L}{2r_H + r_L} < \frac{1}{2}.$$

By Axiom 3, there exists $F \in \mathcal{F}$ such that

$$\mathbf{E}_F[(Q - Q_0)_+] > 0 \quad \text{and} \quad \frac{Q_0 - \mathbf{E}_F[Q]}{\mathbf{E}_F[(Q - Q_0)_+]} \leq \varepsilon.$$

Define

$$Q_L := Q_0 - \mathbf{E}_F[(Q_0 - Q)_+] \quad \text{and} \quad Q_H := 2\mathbf{E}_F[Q] + \mathbf{E}_F[(Q - Q_0)_+] - Q_0.$$

It is easy to see that $0 \leq Q_L < Q_0$; by our earlier choice of ε , $Q_0 < Q_H$. The definition of Q_0 implies that

$$Q_H = Q_0 + \mathbf{E}_F[(Q - Q_0)_+] - 2[Q_0 - \mathbf{E}_F[Q]] \leq Q_0 + \mathbf{E}_F[(Q - Q_0)_+] \leq 1.$$

Thus, we can define the three-tier distribution

$$F^{(3)} := \frac{1}{3}\delta_{Q_L} + \frac{1}{3}\delta_{Q_0} + \frac{1}{3}\delta_{Q_H}.$$

We now show that the distribution $F^{(3)}$ is a mean-preserving contraction of F and hence feasible by Axiom 1. For $z \leq Q_0$, Jensen's inequality implies that

$$\begin{aligned} \mathbf{E}_{F^{(3)}}[(z - Q)_+] &= \frac{1}{3}(z - Q_L)_+ = \frac{1}{3}[z - Q_0 + \mathbf{E}_F[(Q_0 - Q)_+]]_+ \\ &\leq \frac{1}{3}\mathbf{E}_F[(z - Q)_+] \leq \mathbf{E}_F[(z - Q)_+]. \end{aligned}$$

Similarly, for $z > Q_0$,

$$\begin{aligned}\mathbf{E}_{F^{(3)}}[(Q - z)_+] &= \frac{1}{3} (Q_H - z)_+ \leq \frac{1}{3} [Q_0 + \mathbf{E}_F[(Q - Q_0)_+] - z]_+ \\ &\leq \frac{1}{3} \mathbf{E}_F[(Q - z)_+] \leq \mathbf{E}_F[(Q - z)_+].\end{aligned}$$

Since $F^{(3)}$ and F share the same mean, it follows that $F^{(3)}$ is a mean-preserving contraction of F .

Finally, we show that the implementing mechanism strictly improves every agent's utility. By equation (2), its intermediate-tier payment satisfies

$$\begin{aligned}-p_0 &= \frac{r_H - r_L}{3} \mathbf{E}_F[(Q - Q_0)_+] - \frac{2r_H + r_L}{3} [Q_0 - \mathbf{E}_F[Q]] \\ &\geq \frac{\mathbf{E}_F[(Q - Q_0)_+]}{3} [r_H - r_L - (2r_H + r_L) \varepsilon] > 0.\end{aligned}$$

Here, the final inequality follows from our earlier choice of ε . We conclude that $F^{(3)}$ results in a strict Pareto improvement, as claimed. $\square$

The proof of Theorem 2 shows that a strict Pareto improvement can be achieved with three equally sized tiers. The tier masses can thus be fixed independently of the distribution of agents' types.

Equation (2) also characterizes which feasible three-tier distributions with intermediate quality Q_0 yield a strict Pareto improvement.

Corollary 2. *A feasible three-tier distribution $F = \pi_L \delta_{Q_L} + (1 - \pi_L - \pi_H) \delta_{Q_0} + \pi_H \delta_{Q_H}$, where $Q_L < Q_0 < Q_H$, $\pi_L, \pi_H > 0$, and $\pi_L + \pi_H < 1$, yields a strict Pareto improvement over the equity benchmark if and only if*

$$\frac{Q_H - Q_0}{Q_0 - Q_L} > \frac{\pi_L}{\pi_H} \frac{G^{-1}(\pi_L)}{G^{-1}(1 - \pi_H)}.$$

Corollary 2 relates the scope for a Pareto improvement to heterogeneity in agents' rates of substitution. As $\pi_L = \pi_H \downarrow 0$, the right-hand side of the condition converges to $\underline{r}/\bar{r}$, a simple measure of this heterogeneity and hence of the potential gains from sorting.¹² In Section 4.4, we exploit this observation to relax Axiom 3 by comparing the loss in average quality directly with these gains.

¹² This observation is a point of contact with Ater et al. (2026), who emphasize that driver heterogeneity is crucial for assessing the welfare gains of congestion pricing.

4 Applications

This section applies and generalizes the main results of Section 3 to four settings: lane pricing, waiting in line, public provision of goods with heterogeneous quality, and insurance under adverse selection.

4.1 Lane Pricing

We begin with lane pricing, our leading application and primary motivation for the framework. Recall that there are $N \geq 2$ lanes; if a mass m_i of agents is assigned to lane $i \in \{1, 2, \dots, N\}$, each agent in that lane experiences travel time $w(m_i)$. Congestion effects are captured by $w : [0, 1] \rightarrow [\underline{t}, \bar{t}]$, which is continuously differentiable, increasing, and convex, with $0 \leq \underline{t} < \bar{t} < 1$. A travel time t corresponds to effective quality $v(1 - t)$, where v is continuously differentiable, increasing, and concave.

The laissez-faire allocation in this setting provides a natural candidate for the equity benchmark. In the absence of pricing, agents distribute themselves evenly across lanes, resulting in the lane assignment $\mathbf{m}_0 = (1/N, 1/N, \dots, 1/N)$. The corresponding laissez-faire distribution assigns full probability to the quality $Q_0 := v(1 - w(1/N))$.

In this setting, the laissez-faire allocation uniquely maximizes average quality across all feasible allocations, as shown in Section 2. The convexity and strict monotonicity of the congestion technology induce a mean–dispersion tradeoff: any dispersion in qualities requires a departure from equal lane usage, which necessarily decreases average quality. At the same time, this tradeoff is locally mild. Specifically, the decrease in average quality induced by small amounts of dispersion is of higher order, as formally demonstrated in Appendix A.1.

Given that Axioms 1 to 3 hold in this setting, our main results apply: Theorems 1 and 2 imply that there exists a strict Pareto improvement with three tiers, but no Pareto improvement with two tiers. As shown in Appendix A.1, our main results extend to deterministic lane assignments:

Proposition 1. *When $N \geq 3$, there exists a mechanism using only deterministic lane assignments that strictly Pareto-improves on the laissez-faire outcome. When $N = 2$, no mechanism using only deterministic lane assignments can be a Pareto improvement.*

Proposition 1 has a simple practical interpretation. With at least three physical lanes, the Pareto improvement can be implemented deterministically: commuters can be assigned to fast, regular, and slow lanes, with no need to randomize their access. Prices can then be calibrated so that the regular lane has the same level of congestion as in the laissez-faire allocation.

With only two physical lanes, some randomization is necessary to generate a Pareto improvement, but this need not require literal lotteries. It could instead be implemented through restrictions on access over time—for example, by allowing commuters choosing the intermediate tier to use the fast lane only on certain days or for a limited amount of time.

Our positive results rely on the ability to redistribute toll revenues to commuters. As emphasized in the congestion-pricing literature (e.g., Hall, 2018), such redistribution may be difficult in practice. Importantly, however, our proof of Theorem 2 implies that redistribution need not be perfect. Fixing the distribution G of commuters' types, there exists a cutoff $\bar{\alpha} < 1$ such that a Pareto improvement remains possible whenever a fraction $\alpha > \bar{\alpha}$ of toll revenues can be returned to commuters. Such redistribution could be implemented indirectly, for example through reductions in other taxes or fees paid by drivers, or through credits deposited into electronic toll accounts that can ultimately be redeemed for cash.¹³

4.2 Waiting in Line

We next consider a designer who can *endogenously create lines* by partitioning agents and dedicating processing capacity to each line, as in airport security screening or access to public services. Unlike our lane-pricing application, where congestion depends only on the mass assigned to each lane, here congestion depends on demand relative to capacity in each line.

In this setting, agents arrive at a constant rate $\lambda > 0$, and the designer has total processing capacity $\mu > 0$. Denote the demand–capacity ratio by $\rho := \lambda/\mu \in (0, 1)$, which we refer to as the *load* of the entire system. A deterministic line design consists of a finite number of lines $K \in \{1, 2, \dots\}$ together with routing shares $\mathbf{m} = (m_1, \dots, m_K) \in \Delta^{K-1}$ and capacity shares $\mathbf{s} = (s_1, \dots, s_K) \in \Delta^{K-1}$. Given such

¹³ What our framework abstracts from is the possible extensive-margin responses to the introduction of priority pricing. By definition, a strict Pareto improvement among existing commuters increases the attractiveness of driving and may therefore induce additional traffic.

a design $(K, \mathbf{m}, \mathbf{s})$, line $i \in \{1, 2, \dots, K\}$ is assigned arrival rate λm_i and capacity μs_i , so its load is

$$\rho_i := \frac{\lambda m_i}{\mu s_i} = \rho \frac{m_i}{s_i}.$$

We restrict attention to designs in which $m_i, s_i > 0$ and $\rho_i < 1$ for every i .

We model delay using a reduced-form congestion technology. Each agent assigned to line i faces the deterministic waiting time $w(\rho_i)$, where $w : (0, 1) \rightarrow [\underline{t}, \bar{t}]$ is continuously differentiable, increasing, and convex, with $0 \leq \underline{t} < \bar{t} < 1$. The key property of this technology is *scale-freeness*: if both demand and capacity in a line are scaled by the same factor, the load ρ_i is unchanged and so is the wait time. As in the lane-pricing application, the quality associated with wait time $w(\rho_i)$ is $v(1 - w(\rho_i))$.

Next, we specify the feasible set of expected quality distributions. We identify each deterministic line design $(K, \mathbf{m}, \mathbf{s})$ with the quality distribution that it induces,

$$F_{(K, \mathbf{m}, \mathbf{s})} := \sum_{i=1}^K m_i \delta_{v(1-w(\rho_i))}.$$

The designer can randomize agents' assignments within a deterministic line design, so the feasible set is

$$\mathcal{F} = \{F \in \Delta([0, 1]) : F \in \text{MPC}(F_{(K, \mathbf{m}, \mathbf{s})}) \text{ for some deterministic line design } (K, \mathbf{m}, \mathbf{s})\}.$$

Under the laissez-faire outcome, all agents use a single line. This corresponds to a design with $K = 1$ and $m_1 = s_1 = 1$. Each agent receives quality $Q_0 = v(1 - w(\rho))$, so the induced distribution is δ_{Q_0} .

All three axioms hold in this setting, with the laissez-faire expected quality distribution as the equity benchmark. Axiom 1 holds by construction; Appendix A.2 verifies Axioms 2 and 3 by an argument similar to the lane-pricing application. Intuitively, making one line faster requires allocating more capacity to it relative to demand, leaving another line with less capacity relative to demand. Because waiting time is convex in line load, this dispersion increases average wait time. Moreover, small departures from proportional capacity allocation create first-order dispersion at a higher-order cost to the average.

Since Axioms 1 to 3 hold in this setting, Theorems 1 and 2 apply. As in the lane-pricing application, the results extend to deterministic assignments to lines:

Proposition 2. *There exists a mechanism using only deterministic assignments to three lines that strictly Pareto-improves on the laissez-faire outcome. No mechanism using only deterministic assignments to at most two lines can be a Pareto improvement.*

The implementation suggested by Proposition 2 is again simple. The designer can partition existing capacity across three lines, assigning relatively more capacity per user to the fast line, relatively less to the slow line, and preserving the laissez-faire demand–capacity ratio in the intermediate line. In applications, this can be implemented through separate queues or service windows with different capacity allocations.

4.3 Public Provision With Heterogeneous Quality

Our third application considers environments in which the designer allocates goods of heterogeneous physical quality. It illustrates how the mean–dispersion tradeoff underlying our results can arise from economic frictions unrelated to congestion.

We consider a unit mass of agents who demand a good of heterogeneous physical quality, $q \in [0, 1]$. For example, q might represent the size of a public housing unit. Each agent privately observes his type $r \in [\underline{r}, \bar{r}] \subset \mathbf{R}_+$. An agent of type r who receives physical quality q and pays price p obtains utility $rv(q) - p$, where $v : [0, 1] \rightarrow [0, 1]$ is continuously differentiable, increasing, and strictly concave.

A capacity-constrained designer has $q_0 \in (0, 1)$ units of physical quality to allocate per capita. For instance, q_0 might represent the total amount of space that a public housing authority can subdivide into individual units. Given an allocation function $q : [\underline{r}, \bar{r}] \rightarrow [0, 1]$, feasibility requires that the total physical quality allocated not exceed available capacity:

$$\int_{\underline{r}}^{\bar{r}} q(r) dG(r) \leq q_0.$$

To apply the framework of Section 2, we express this constraint in terms of effective quality. Because $Q = v(q)$ and v is increasing, providing effective quality Q requires physical quality $v^{-1}(Q)$. Accordingly, the feasible set of expected effective quality distributions is

$$\mathcal{F} = \{F \in \Delta([0, v(1)]) : \mathbf{E}_F[v^{-1}(Q)] \leq q_0\}.$$

Because v is concave, any lottery delivering expected effective quality Q uses at least $v^{-1}(Q)$ units of expected physical quality; assigning physical quality $v^{-1}(Q)$ with certainty attains this bound.

In the absence of pricing, the designer cannot use quality to screen agents; a natural candidate for the equity benchmark is therefore uniform provision at physical quality q_0 . This allocation provides every agent with effective quality $Q_0 := v(q_0)$.

The feasible set satisfies all three axioms. Axiom 1 follows because v^{-1} is convex, so the capacity constraint is preserved under mean-preserving contractions. We verify Axiom 2 below and Axiom 3 in Appendix A.3. For any feasible distribution F , Jensen's inequality gives

$$\mathbf{E}_F[Q] = \mathbf{E}_F\left[v\left(v^{-1}(Q)\right)\right] \leq v\left(\mathbf{E}_F\left[v^{-1}(Q)\right]\right) \leq v(q_0) = Q_0.$$

Because v is strictly concave, equality holds if and only if physical quality is almost surely constant at q_0 . Thus, the unique maximizing feasible distribution is δ_{Q_0} . Any dispersion in physical quality around the equity benchmark necessarily decreases average *effective* quality due to decreasing marginal utility from physical quality, yielding the key mean–dispersion tradeoff formalized by Axiom 2.

Since Axioms 1 to 3 hold in this public-provision environment, our main results apply:

Proposition 3. *Any Pareto improvement over the equity benchmark must offer at least three levels of expected quality. Moreover, there exists a strict Pareto improvement that offers three levels of expected quality.*

Public housing provides a natural interpretation of this result. Public housing programs often restrict households to a relatively limited set of options, making uniform provision a natural benchmark (e.g., Olsen, 2003; Sitaraman and Alstott, 2019). Singapore's public housing system provides a useful contrast: the public housing authority chooses among different types of apartments to build and allocate, allowing households to express preferences over these alternatives (e.g., Ferdowsian, Lee and Yap, 2026). Our results suggest that differentiation—in particular, in size—need not come at the expense of equity. If prices are appropriately adjusted, offering at least three quality levels can generate a Pareto improvement over uniform provision, even if dispersion in physical quality lowers average effective quality (e.g., due to decreasing marginal utility over size).

4.4 Insurance Under Adverse Selection

Our final application shows how our main results generalize beyond the axioms introduced in Section 2. We consider an insurance program with a fixed budget, where the main friction generating the mean–dispersion tradeoff arises from adverse selection. Employer-provided health insurance (e.g., [Einav, Finkelstein and Cullen, 2010](#)) provides a natural example: a self-insured employer with a fixed benefits budget may offer plans of different generosity to employees whose willingness to pay for coverage is positively correlated with their expected medical costs.

Unlike the preceding applications, the resource cost of coverage depends on the type of the agent who receives it due to adverse selection. We first argue that the feasible set satisfies Axioms 1 and 2. We then show that Axiom 3 does not hold. Finally, we provide a more general condition under which a three-tier Pareto improvement nevertheless exists.

Let $Q \in [0, 1]$ denote the generosity of insurance coverage. A type- r agent who receives coverage Q and makes a premium adjustment p obtains utility $rQ - p$. Providing one unit of coverage to this agent generates expected cost $c(r)$ to the designer, where $c : [\underline{r}, \bar{r}] \rightarrow \mathbf{R}_{++}$ is continuous and increasing. Thus, willingness to pay and expected claims are positively associated, as in a standard adverse-selection environment. A simple example is when r reflects the probability of using the insurance, while $c(r)$ also incorporates the expected expenditure conditional on use.

The equity benchmark provides uniform coverage $Q_0 \in (0, 1)$ with no premium adjustments. An allocation $Q : [\underline{r}, \bar{r}] \rightarrow [0, 1]$ is feasible if the aggregate expected cost does not exceed that under the equity benchmark:

$$\int_{\underline{r}}^{\bar{r}} c(r)Q(r) dG(r) \leq Q_0 \int_{\underline{r}}^{\bar{r}} c(r) dG(r).$$

Premium adjustments must satisfy budget balance. To express the feasibility constraint in terms of distributions of expected quality, define $C(u) := c(G^{-1}(u))$ for every $u \in [0, 1]$. Because incentive compatibility assigns higher coverage to higher types, an expected quality distribution F results in

$$\int_{\underline{r}}^{\bar{r}} c(r)Q(r) dG(r) = \int_0^1 C(u)F^{-1}(u) du$$

units of expected cost. Accordingly, the feasible set is

$$\mathcal{F} = \left\{ F \in \Delta([0, 1]) : \int_0^1 C(u)F^{-1}(u) du \leq Q_0 \int_0^1 C(u) du \right\}.$$

We first verify Axiom 1. For any $\bar{F} \in \text{MPC}(F)$, since dC is a nonnegative measure under our maintained assumptions, integration by parts yields

$$\begin{aligned} \int_0^1 C(u)\bar{F}^{-1}(u) du &= C(1) \int_0^1 \bar{F}^{-1}(z) dz - \int_0^1 \int_0^u \bar{F}^{-1}(z) dz dC(u) \\ &\leq C(1) \int_0^1 F^{-1}(z) dz - \int_0^1 \int_0^u F^{-1}(z) dz dC(u) = \int_0^1 C(u)F^{-1}(u) du. \end{aligned}$$

Thus, every mean-preserving contraction of a feasible distribution remains feasible, and Axiom 1 holds.

Next, we verify Axiom 2. Since both C and F^{-1} are nondecreasing, Chebyshev's integral inequality gives

$$\int_0^1 C(u)F^{-1}(u) du \geq \left[\int_0^1 C(u) du \right] \left[\int_0^1 F^{-1}(u) du \right] = \mathbf{E}_F[Q] \int_0^1 C(u) du.$$

Feasibility therefore implies that $\mathbf{E}_F[Q] \leq Q_0$. Because C is increasing, this inequality is strict whenever F is nondegenerate. Hence, the distribution δ_{Q_0} uniquely maximizes average quality, and Axiom 2 holds.

Unlike in the preceding applications, however, the mean–dispersion tradeoff is not locally mild, due to adverse selection. Increasing coverage for high types is more expensive than reducing the same amount of coverage for low types saves, so feasibility requires a non-negligible additional reduction in coverage at the bottom. As a result, the loss in average quality cannot be of lower order than the quality gains at the top. In Appendix A.4, we show that Axiom 3 fails.

Even without Axiom 3, however, a three-tier strict Pareto improvement remains possible if high types value coverage sufficiently highly relative to their expected cost. We consider the condition

$$\frac{\bar{r}}{c(\bar{r})} > \frac{\underline{r}}{c(\underline{r})}. \quad (\text{B})$$

The “bang-for-the-buck” ratio $r/c(r)$ measures a type's willingness to pay for coverage per unit of expected cost. While this ratio may vary nonmonotonically with type, condition (B) requires this ratio to

be higher at the upper endpoint than at the lower endpoint.

Proposition 4. *There does not exist a two-tier Pareto improvement over uniform coverage. However, under condition (B), there exists a three-tier strict Pareto improvement.*

In the context of employer-provided health insurance, Proposition 4 implies that—with a fixed benefits budget and budget-balanced premium adjustments—offering a choice between a low-coverage and a high-coverage plan cannot make all employees better off compared to a simple uniform-coverage scheme. A third, intermediate-coverage plan can achieve a Pareto improvement when employees with the highest willingness to pay for coverage have sufficiently high willingness to pay relative to their expected claims.

Without some restriction on the relationship between willingness to pay and expected cost, a Pareto improvement need not exist. Suppose, for example, that $\underline{r} > 0$ and $c(r) = r$. For any feasible, budget-balanced mechanism,

$$\int_{\underline{r}}^{\bar{r}} [rQ(r) - p(r) - rQ_0] dG(r) = \int_{\underline{r}}^{\bar{r}} r [Q(r) - Q_0] dG(r) = \int_{\underline{r}}^{\bar{r}} c(r) [Q(r) - Q_0] dG(r) \leq 0.$$

A Pareto improvement would make the first integrand nonnegative for every type and strictly positive for a positive mass of types, contradicting this inequality. Thus, no Pareto improvement exists in this case, regardless of the number of tiers.

Condition (B) motivates a generalization of Axiom 3. While our framework in Section 2 deliberately abstracted from variable production costs for simplicity, such costs can be accommodated:

Axiom 3* (Generalized Smoothness). *There exists a continuous function $\kappa : [0, 1] \rightarrow \mathbf{R}_{++}$ such that:*

- (i) *For every bounded nondecreasing function $h : [0, 1] \rightarrow \mathbf{R}$ satisfying $\int_0^1 \kappa(u)h(u) du < 0$, there exists a sequence $\{(\varepsilon_n, F_n)\}_{n=1}^\infty$ such that $\varepsilon_n \downarrow 0$, $F_n \in \mathcal{F}$, and*

$$\int_0^1 \left| \frac{F_n^{-1}(u) - Q_0}{\varepsilon_n} - h(u) \right| du \rightarrow 0.$$

- (ii) *The extreme types satisfy*

$$\frac{G^{-1}(1)}{\kappa(1)} > \frac{G^{-1}(0)}{\kappa(0)}.$$

Theorem 2^{*}. *Under Axioms 1 and 3^{*}, there exists a strict Pareto improvement with three tiers.*

Proof. See Appendix B. □

Intuitively, Axiom 3^{*} replaces the requirement that the loss in average quality be of higher order with a condition on the local resource tradeoff. The function $\kappa(u)$ can be interpreted as the marginal resource cost of increasing expected quality at quantile u . Condition (i) requires every bounded, nondecreasing perturbation that strictly reduces resource use to be approximated by arbitrarily small feasible departures from the equity benchmark. Condition (ii) requires the highest type to have a higher willingness to pay per unit of marginal resource cost than the lowest type. Together, these conditions make it possible to reduce quality for a small mass of low types and increase quality for a small mass of high types, while retaining Q_0 as the intermediate tier. The resulting gains can then be redistributed through payments so that every type is strictly better off.

In our insurance application, condition (i) of Axiom 3^{*} holds with $\kappa = C$, while condition (ii) is equivalent to condition (B). This shows that Axiom 3^{*} can hold even when Axiom 3 fails. In contrast, under Axioms 1 and 2, Axiom 3 implies Axiom 3^{*} (see Appendix B).

5 Concluding Remarks

This paper shows that, under economically natural assumptions, three priority tiers are necessary and sufficient for a Pareto improvement over the equity benchmark. With only two tiers, the agent indifferent between them receives no sorting gain to offset the loss in average quality and is therefore worse off. A third tier separates the threshold between the low and intermediate tiers from the threshold between the intermediate and high tiers. This separation allows sorting gains to outweigh the loss in average quality for every agent.

Our framework isolates this logic without specifying interpersonal welfare comparisons or committing to a particular allocation technology. It applies to lane pricing, waiting-line design, and public provision, and extends to insurance under adverse selection, where the resource cost of quality depends on the recipient. More broadly, our results show that the distributional consequences of priority pricing depend

not only on whether prices are introduced, but also on how the priority tiers are designed. Policy debates framed as a choice between uniform access and a two-tier priority system therefore overlook a simple alternative: a third tier can allow the gains from priority pricing to be shared by all agents.

References

- AKBARPOUR, M. ® P. DWORCZAK ® S. D. KOMINERS (2024): “Redistributive Allocation Mechanisms,” *Journal of Political Economy*, 132, 1831–1875.
- ANDERSON, M. G. (2005): “The Bias Built Into ‘Lexus Lanes’,” *The Washington Post*, <https://www.washingtonpost.com/archive/opinions/2005/07/01/the-bias-built-into-lexus-lanes/8fa5656e-83b4-47c8-af11-f6d3fb08c661/>, Letter to the Editor.
- ARNOTT, R., A. DE PALMA, AND R. LINDSEY (1994): “The Welfare Effects of Congestion Tolls With Heterogeneous Commuters,” *Journal of Transport Economics and Policy*, 28, 139–161.
- ATER, I., B. ROSS, A. SHANY, AND S. VASSERMAN (2026): “Road Pricing Under Heterogeneity,” *Working paper*.
- BEHESHTIAN, A., R. R. GEDDES, O. M. ROUHANI, K. M. KOCKELMAN, A. OCKENFELS, P. CRAMTON, AND W. DO (2020): “Bringing the Efficiency of Electricity Market Mechanisms to Multimodal Mobility Across Congested Transportation Systems,” *Transportation Research Part A: Policy and Practice*, 131, 58–69.
- VAN DEN BERG, V., AND E. T. VERHOEF (2011): “Congestion Tolling in the Bottleneck Model With Heterogeneous Values of Time,” *Transportation Research Part B: Methodological*, 45, 60–78.
- BOBBIO, E., P. CRAMTON, AND A. OCKENFELS (2021): “Welfare Effects of Efficient Road Pricing in a Simple Model of a City With Heterogeneous Agents,” *Working paper*.
- CHAO, H.-P., AND R. B. WILSON (1987): “Priority Service: Pricing, Investment, and Market Organization,” *American Economic Review*, 77, 899–916.

- CONDORELLI, D. (2013): “Market and Non-Market Mechanisms for the Optimal Allocation of Scarce Resources,” *Games and Economic Behavior*, 82, 582–591.
- COOK, C., A. KREIDIEH, S. VASSERMAN, H. ALLCOTT, N. ARORA, A. TOMKINS, E. TURKEL, AND F. VAN SAMBEEK (2026): “The Network-Wide Effects of Congestion Pricing: Evidence From New York City,” *Working paper*.
- COOK, C., AND P. Z. LI (2025): “Value Pricing or Lexus Lanes? The Distributional Effects of Dynamic Tolling,” *Working paper*.
- CRAMTON, P., R. R. GEDDES, AND A. OCKENFELS (2019): “Markets for Road Use: Eliminating Congestion Through Scheduling, Routing, and Real-Time Road Pricing,” *Working paper*.
- DWORCZAK, P. & S. D. KOMINERS & M. AKBARPOUR (2021): “Redistribution Through Markets,” *Econometrica*, 89, 1665–1698.
- EINAV, L., A. FINKELSTEIN, AND M. R. CULLEN (2010): “Estimating Welfare in Insurance Markets Using Variation in Prices,” *Quarterly Journal of Economics*, 125, 877–921.
- FERDOWSIAN, A., K. H. LEE, AND L. YAP (2026): “Build-To-Order: Endogenous Supply in Centralized Mechanisms,” *Working paper*.
- GERSHKOV, A., AND P. SCHWEINZER (2010): “When Queueing Is Better Than Push and Shove,” *International Journal of Game Theory*, 39, 409–430.
- GERSHKOV, A., AND E. WINTER (2023): “Gainers and Losers in Priority Services,” *Journal of Political Economy*, 131, 3103–3155.
- HALL, J. D. (2018): “Pareto Improvements From Lexus Lanes: The Effects of Pricing a Portion of the Lanes on Congested Highways,” *Journal of Public Economics*, 158, 113–125.
- (2021): “Can Tolling Help Everyone? Estimating the Aggregate and Distributional Consequences of Congestion Pricing,” *Journal of the European Economic Association*, 19, 441–474.

- KANG, Z. Y. (2023): “The Public Option and Optimal Redistribution,” *Working paper*.
- KANG, Z. Y., AND M. WATT (2026): “Topping Up and Optimal Redistribution,” *Working paper*.
- KREINDLER, G. (2024): “Peak-Hour Road Congestion Pricing: Experimental Evidence and Equilibrium Implications,” *Econometrica*, 92, 1233–1268.
- MILGROM, P., AND I. SEGAL (2002): “Envelope Theorems for Arbitrary Choice Sets,” *Econometrica*, 70, 583–601.
- MULL, A. (2023): “The Perfect Service to Make Everyone at the Airport Hate You,” *The Atlantic*, <https://www.theatlantic.com/technology/archive/2023/07/clear-airport-security-lines-tsa-infrastructure/674809/>.
- MÜLLER, A., AND D. STOYAN (2002): *Comparison Methods for Stochastic Models and Risks*, Volume 389, West Sussex, UK: John Wiley & Sons.
- MYERSON, R. B. (1981): “Optimal Auction Design,” *Mathematics of Operations Research*, 6, 58–73.
- NICHOLS, A. L., AND R. J. ZECKHAUSER (1982): “Targeting Transfers through Restrictions on Recipients,” *American Economic Review*, 72, 372–377.
- OLSEN, E. O. (2003): “Housing Programs for Low-Income Households,” in *Means-Tested Transfer Programs in the United States* ed. by Moffitt, R. A.: University of Chicago Press.
- OSTROVSKY, M., AND M. SCHWARZ (2018): “Carpooling and the Economics of Self-Driving Cars,” *Working paper*.
- OSTROVSKY, M., AND F. YANG (2024): “Effective and Equitable Congestion Pricing: New York City and Beyond,” *Working paper*.
- PIGOU, A. C. (1920): *The Economics of Welfare*, London, UK: Macmillan.
- SHACHAR, A. (2018): “Dangerous Liaisons: Money and Citizenship,” in *Debating Transformations of National Citizenship* ed. by Bauböck, R. Cham: Springer, 7–15, [10.1007/978-3-319-92719-0_2](https://doi.org/10.1007/978-3-319-92719-0_2).

- SITARAMAN, G., AND A. L. ALSTOTT (2019): *The Public Option: How to Expand Freedom, Increase Opportunity, and Promote Equality*, Cambridge, MA: Harvard University Press.
- SPENCE, M. (1977): “Nonlinear Prices and Welfare,” *Journal of Public Economics*, 8, 1–18.
- STEWART, E. (2022): “Why Must We Pay to Have a Slightly Less Miserable Time at the Airport?” *Vox*, <https://www.vox.com/the-goods/23101906/tsa-precheck-clear-cost-airport-security-travel>.
- VICKREY, W. S. (1969): “Congestion Theory and Transport Investment,” *American Economic Review (Papers and Proceedings)*, 59, 251–260.
- WEITZMAN, M. L. (1977): “Is the Price System or Rationing More Effective in Getting a Commodity to Those Who Need It Most?” *Bell Journal of Economics*, 8, 517–524.

A Omitted Proofs

A.1 Proofs for Section 4.1

We first argue that Axiom 3 holds in this application. Consider a perturbation of the laissez-faire allocation. For $\varepsilon \in (0, 1/N)$, define

$$\mathbf{m}_\varepsilon := \left(\frac{1}{N} - \varepsilon, \frac{1}{N}, \dots, \frac{1}{N}, \frac{1}{N} + \varepsilon \right),$$

so that a mass ε of agents is shifted from lane 1 to lane N , while all other lanes retain their laissez-faire masses. Writing F^ε for the induced distribution of qualities, the resulting loss in average quality is

$$Q_0 - \mathbf{E}_{F^\varepsilon}[Q] = 2\phi\left(\frac{1}{N}\right) - \phi\left(\frac{1}{N} - \varepsilon\right) - \phi\left(\frac{1}{N} + \varepsilon\right) = o(\varepsilon).$$

The final equality follows from the differentiability of ϕ : the two first-order effects cancel. By contrast, the increase in dispersion—measured by the cumulative improvement above Q_0 received by agents in lane 1—is

$$\mathbf{E}_{F^\varepsilon}[(Q - Q_0)_+] = \left(\frac{1}{N} - \varepsilon\right) \left[v\left(1 - w\left(\frac{1}{N} - \varepsilon\right)\right) - v\left(1 - w\left(\frac{1}{N}\right)\right) \right] = \frac{\varepsilon}{N} v'\left(1 - w\left(\frac{1}{N}\right)\right) w'\left(\frac{1}{N}\right) + o(\varepsilon).$$

Because $v', w' > 0$, this expression is positive for all sufficiently small ε and is of order ε . Hence, Axiom 3 holds:

$$\lim_{\varepsilon \downarrow 0} \frac{Q_0 - \mathbf{E}_{F^\varepsilon}[Q]}{\mathbf{E}_{F^\varepsilon}[(Q - Q_0)_+]} = 0.$$

We now prove Proposition 1. Suppose first that $N = 2$. Any deterministic lane assignment induces a quality distribution with at most two tiers. A one-tier distribution cannot be a Pareto improvement: incentive compatibility requires all payments to be equal, budget balance requires them to be zero, and feasibility implies that the common quality is no greater than Q_0 . A two-tier distribution cannot be a Pareto improvement by Theorem 1, since Axiom 2 holds in this setting. Hence, no mechanism using only deterministic lane assignments can be a Pareto improvement when $N = 2$.

Now suppose that $N \geq 3$. For $\varepsilon \in (0, 1/N)$, consider the deterministic lane assignment

$$\mathbf{m}_\varepsilon := \left(\frac{1}{N} - \varepsilon, \frac{1}{N}, \dots, \frac{1}{N}, \frac{1}{N} + \varepsilon \right).$$

The induced distribution F^ε has three tiers. Writing

$$Q_L := v\left(1 - w\left(\frac{1}{N} + \varepsilon\right)\right) \quad \text{and} \quad Q_H := v\left(1 - w\left(\frac{1}{N} - \varepsilon\right)\right),$$

these tiers are $Q_L < Q_0 < Q_H$, with respective masses $1/N + \varepsilon$, $(N - 2)/N$, and $1/N - \varepsilon$.

Consider the mechanism implementing F^ε . The threshold types between the low and intermediate tiers and between the intermediate and high tiers are

$$r_L = G^{-1}\left(\frac{1}{N} + \varepsilon\right) \quad \text{and} \quad r_H = G^{-1}\left(1 - \frac{1}{N} + \varepsilon\right).$$

By equation (2), the intermediate-tier payment p_0 satisfies

$$-p_0 = \left(\frac{1}{N} - \varepsilon\right) r_H (Q_H - Q_0) - \left(\frac{1}{N} + \varepsilon\right) r_L (Q_0 - Q_L).$$

As $\varepsilon \downarrow 0$,

$$\begin{cases} Q_H - Q_0 &= v'\left(1 - w\left(\frac{1}{N}\right)\right) w'\left(\frac{1}{N}\right) \varepsilon + o(\varepsilon), \\ Q_0 - Q_L &= v'\left(1 - w\left(\frac{1}{N}\right)\right) w'\left(\frac{1}{N}\right) \varepsilon + o(\varepsilon). \end{cases}$$

Therefore,

$$-p_0 = \frac{\varepsilon}{N} v'\left(1 - w\left(\frac{1}{N}\right)\right) w'\left(\frac{1}{N}\right) \left[G^{-1}\left(1 - \frac{1}{N}\right) - G^{-1}\left(\frac{1}{N}\right) \right] + o(\varepsilon).$$

Because $N \geq 3$, we have $1 - 1/N > 1/N$. Since G^{-1} is strictly increasing and $v', w' > 0$, it follows that $-p_0 > 0$ for all sufficiently small $\varepsilon > 0$.

By the argument in Section 3.2, the positive intermediate-tier subsidy implies that the mechanism strictly Pareto-improves on the laissez-faire outcome for all sufficiently small $\varepsilon > 0$.

A.2 Proofs for Section 4.2

We first prove that Axioms 2 and 3 hold. As in our lane-pricing application, since randomization does not affect average effective quality, it suffices to show that every deterministic line design has average effective quality at most Q_0 and that every design attaining Q_0 induces the distribution δ_{Q_0} . To this end, define $\phi(y) := yv(1 - w(\rho y))$; since w is increasing and convex on $(0, 1)$, it follows that $\phi : (0, 1/\rho) \rightarrow \mathbf{R}$ is strictly concave. For any deterministic line design $(K, \mathbf{m}, \mathbf{s})$, average effective quality can be written as

$$\sum_{i=1}^K s_i \phi\left(\frac{m_i}{s_i}\right) \leq \phi\left(\sum_{i=1}^K s_i \frac{m_i}{s_i}\right) = \phi(1) = v(1 - w(\rho)) = Q_0.$$

Since ϕ is strictly concave, equality holds if and only if m_i/s_i is constant across lines. Because $\sum_{i=1}^K m_i = \sum_{i=1}^K s_i = 1$, this constant must be 1. Hence, equality holds if and only if $m_i = s_i$ for every i , in which case every line has load ρ and the induced distribution is δ_{Q_0} . Thus, Axiom 2 holds.

Axiom 3 is also satisfied in this setting. To see this, fix $\eta \in (0, 1/2)$. For any $\varepsilon > 0$ sufficiently small that $\rho(1 + \varepsilon) < 1$, consider a deterministic three-line design with capacity and routing shares

$$(s^F, s^0, s^S) = (\eta, 1 - 2\eta, \eta) \quad \text{and} \quad (m^F, m^0, m^S) = (\eta(1 - \varepsilon), 1 - 2\eta, \eta(1 + \varepsilon)).$$

The corresponding loads are $\rho(1 - \varepsilon)$, ρ , and $\rho(1 + \varepsilon)$, so the induced distribution F_ε has three support points

$$v(1 - w(\rho(1 - \varepsilon))) > Q_0 = v(1 - w(\rho)) > v(1 - w(\rho(1 + \varepsilon))).$$

These have corresponding masses $\eta(1 - \varepsilon)$, $1 - 2\eta$, and $\eta(1 + \varepsilon)$. By construction, the loss in average effective quality relative to the equity benchmark is

$$Q_0 - \mathbf{E}_{F_\varepsilon}[Q] = \eta[2\phi(1) - \phi(1 - \varepsilon) - \phi(1 + \varepsilon)] = o(\varepsilon).$$

By contrast, the cumulative improvement above Q_0 received by agents in the fast line is

$$\mathbf{E}_{F_\varepsilon}[(Q - Q_0)_+] = \eta(1 - \varepsilon)[v(1 - w(\rho(1 - \varepsilon))) - v(1 - w(\rho))] = \eta\rho v'(1 - w(\rho))w'(\rho)\varepsilon + o(\varepsilon).$$

Because $v', w' > 0$, this expression is positive for all sufficiently small ε and is of order ε . Consequently, Axiom 3 holds:

$$\lim_{\varepsilon \downarrow 0} \frac{Q_0 - \mathbf{E}_{F_\varepsilon}[Q]}{\mathbf{E}_{F_\varepsilon}[(Q - Q_0)_+]} = 0.$$

We now sketch the proof of Proposition 2, which closely parallels the proof of Proposition 1.

No mechanism using only deterministic assignments to at most two lines can be a Pareto improvement. Such assignments induce at most two levels of expected quality. A one-tier distribution cannot be a Pareto improvement by the argument in Section 3; a two-tier distribution cannot be a Pareto improvement by Theorem 1, since Axiom 2 holds in this setting.

Now suppose $K = 3$. Fix $\eta \in (0, 1/2)$ and, for $\varepsilon > 0$ sufficiently small that $\rho(1 + \varepsilon) < 1$, consider the deterministic three-line design

$$(s^F, s^0, s^S) = (\eta, 1 - 2\eta, \eta) \quad \text{and} \quad (m^F, m^0, m^S) = (\eta(1 - \varepsilon), 1 - 2\eta, \eta(1 + \varepsilon)),$$

already used above to verify Axiom 3. As shown there, the induced distribution F_ε assigns mass $\pi_H := \eta(1 - \varepsilon)$ to $Q_H := v(1 - w(\rho(1 - \varepsilon)))$, mass $1 - 2\eta$ to Q_0 , and mass $\pi_L := \eta(1 + \varepsilon)$ to $Q_L := v(1 - w(\rho(1 + \varepsilon)))$, with $Q_L < Q_0 < Q_H$.

Consider the mechanism implementing F_ε . The threshold types between the slow and intermediate lines and between the intermediate and fast lines are

$$r_L := G^{-1}(\eta(1 + \varepsilon)) \quad \text{and} \quad r_H := G^{-1}(1 - \eta(1 - \varepsilon)).$$

By equation (2), the intermediate-tier payment p_0 satisfies

$$-p_0 = \pi_H r_H (Q_H - Q_0) - \pi_L r_L (Q_0 - Q_L).$$

By the same argument as used above,

$$Q_H - Q_0 = \rho v'(1 - w(\rho)) w'(\rho) \varepsilon + o(\varepsilon) \quad \text{and} \quad Q_0 - Q_L = \rho v'(1 - w(\rho)) w'(\rho) \varepsilon + o(\varepsilon).$$

As $\varepsilon \downarrow 0$, $r_L \rightarrow G^{-1}(\eta)$ and $r_H \rightarrow G^{-1}(1 - \eta)$, so

$$-p_0 = \eta \rho v'(1 - w(\rho)) w'(\rho) [G^{-1}(1 - \eta) - G^{-1}(\eta)] \varepsilon + o(\varepsilon).$$

Because $\eta \in (0, 1/2)$, we have $1 - \eta > \eta$; since G^{-1} is strictly increasing and $v', w' > 0$, it follows that $-p_0 > 0$ for all sufficiently small $\varepsilon > 0$.

By the argument in Section 3.2, the positive intermediate-tier subsidy implies that the mechanism strictly Pareto-improves on the laissez-faire outcome for all sufficiently small $\varepsilon > 0$.

A.3 Proofs for Section 4.3

We show that Axiom 3 holds. Fix $m \in (0, 1/2)$ and, for $\varepsilon > 0$ sufficiently small, consider a three-tier allocation that assigns mass m to physical quality $q_0 + \varepsilon$, mass m to physical quality $q_0 - \varepsilon$, and the remaining mass to physical quality q_0 . This perturbation preserves average physical quality and is therefore feasible. Writing the induced distribution of effective qualities as

$$F_\varepsilon = m \delta_{v(q_0 - \varepsilon)} + (1 - 2m) \delta_{v(q_0)} + m \delta_{v(q_0 + \varepsilon)},$$

the corresponding loss in average effective quality is

$$Q_0 - \mathbf{E}_{F_\varepsilon}[Q] = m [2v(q_0) - v(q_0 - \varepsilon) - v(q_0 + \varepsilon)] = o(\varepsilon).$$

However, the cumulative improvement above Q_0 received by the mass m of agents assigned physical quality $q_0 + \varepsilon$ is

$$\mathbf{E}_{F_\varepsilon}[(Q - Q_0)_+] = m [v(q_0 + \varepsilon) - v(q_0)] = m v'(q_0) \varepsilon + o(\varepsilon).$$

Consequently, Axiom 3 holds, and Proposition 3 follows from Theorems 1 and 2:

$$\lim_{\varepsilon \downarrow 0} \frac{Q_0 - \mathbf{E}_{F_\varepsilon}[Q]}{\mathbf{E}_{F_\varepsilon}[(Q - Q_0)_+]} = 0.$$

A.4 Proofs for Section 4.4

We first show that the setting of Section 4.4 does not satisfy Axiom 3. For any feasible F such that $\mathbf{E}_F[(Q - Q_0)_+] > 0$, feasibility implies that both $\Pr_F[Q < Q_0]$ and $\Pr_F[Q > Q_0]$ are positive. Chebyshev's integral inequality then implies that

$$\begin{cases} \int_{1-\Pr_F[Q > Q_0]}^1 C(u) [F^{-1}(u) - Q_0] du & \geq \frac{\mathbf{E}_F[(Q - Q_0)_+]}{\Pr_F[Q > Q_0]} \int_{1-\Pr_F[Q > Q_0]}^1 C(u) du, \\ \int_0^{\Pr_F[Q < Q_0]} C(u) [Q_0 - F^{-1}(u)] du & \leq \frac{\mathbf{E}_F[(Q_0 - Q)_+]}{\Pr_F[Q < Q_0]} \int_0^{\Pr_F[Q < Q_0]} C(u) du. \end{cases}$$

Feasibility requires

$$\begin{aligned} \int_{1-\Pr_F[Q > Q_0]}^1 C(u) [F^{-1}(u) - Q_0] du & \leq \int_0^{\Pr_F[Q < Q_0]} C(u) [Q_0 - F^{-1}(u)] du \\ \implies \frac{\mathbf{E}_F[(Q_0 - Q)_+]}{\Pr_F[Q < Q_0]} \int_0^{\Pr_F[Q < Q_0]} C(u) du & \geq \frac{\mathbf{E}_F[(Q - Q_0)_+]}{\Pr_F[Q > Q_0]} \int_{1-\Pr_F[Q > Q_0]}^1 C(u) du. \end{aligned}$$

Consequently,

$$\frac{Q_0 - \mathbf{E}_F[Q]}{\mathbf{E}_F[(Q - Q_0)_+]} = \frac{\mathbf{E}_F[(Q_0 - Q)_+]}{\mathbf{E}_F[(Q - Q_0)_+]} - 1 \geq \frac{\Pr_F[Q < Q_0] \int_{1-\Pr_F[Q > Q_0]}^1 C(u) du}{\Pr_F[Q > Q_0] \int_0^{\Pr_F[Q < Q_0]} C(u) du} - 1.$$

The expression on the right is the ratio of the average of C over the uppermost $\Pr_F[Q > Q_0]$ quantiles to its average over the lowermost $\Pr_F[Q < Q_0]$ quantiles, minus one. Moreover, this lower bound is uniform. Viewed as a function of the two probabilities, the expression on the right extends continuously to the compact set of pairs in $[0, 1]^2$ whose sum is at most one, where the lower and upper averages at zero equal $C(0)$ and $C(1)$, respectively. It is strictly positive throughout this set because C is strictly increasing, and so it has a strictly positive minimum. Thus, Axiom 3 does not hold in this setting.

We now prove Proposition 4.

Because Axiom 2 holds, Theorem 1 immediately implies that there does not exist a two-tier Pareto improvement over uniform coverage.

We next establish the positive result. Under condition (B),

$$\lim_{m \downarrow 0} \frac{G^{-1}(1-m)}{\frac{1}{m} \int_{1-m}^1 C(u) du} = \frac{\bar{r}}{c(\bar{r})} > \frac{\underline{r}}{c(\underline{r})} = \lim_{m \downarrow 0} \frac{G^{-1}(m)}{\frac{1}{m} \int_0^m C(u) du}.$$

Hence, by continuity, there exists $m \in (0, 1/2)$ such that

$$\frac{G^{-1}(1-m)}{\frac{1}{m} \int_{1-m}^1 C(u) du} > \frac{G^{-1}(m)}{\frac{1}{m} \int_0^m C(u) du}.$$

Define

$$Q_L := Q_0 - \frac{\varepsilon}{m} \int_{1-m}^1 C(u) du \quad \text{and} \quad Q_H := Q_0 + \frac{\varepsilon}{m} \int_0^m C(u) du.$$

For $\varepsilon > 0$ sufficiently small, consider the three-tier distribution

$$F^\varepsilon := m\delta_{Q_L} + (1-2m)\delta_{Q_0} + m\delta_{Q_H}.$$

Because $Q_0 \in (0, 1)$, all three support points lie in $[0, 1]$ and are distinct when ε is sufficiently small.

This distribution is feasible. Indeed, relative to uniform coverage, its aggregate expected cost is

$$\begin{aligned} \int_0^1 C(u) [(F^\varepsilon)^{-1}(u) - Q_0] du &= -\frac{\varepsilon}{m} \left[\int_0^m C(u) du \right] \left[\int_{1-m}^1 C(u) du \right] \\ &\quad + \frac{\varepsilon}{m} \left[\int_{1-m}^1 C(u) du \right] \left[\int_0^m C(u) du \right] = 0. \end{aligned}$$

The implementing mechanism strictly Pareto-improves on uniform coverage. Let $r_L = G^{-1}(m)$ and $r_H = G^{-1}(1-m)$ denote the two threshold types. By equation (2) and our earlier choice of m , the intermediate-tier premium adjustment p_0 satisfies

$$-p_0 = mr_H(Q_H - Q_0) - mr_L(Q_0 - Q_L) = \varepsilon \left[r_H \int_0^m C(u) du - r_L \int_{1-m}^1 C(u) du \right] > 0.$$

By the argument in Section 3.2, the positive intermediate-tier subsidy implies that F^ε yields a three-tier strict Pareto improvement over uniform coverage.

B Additional Results and Generalizations

B.1 Proof of Theorem 2*

By condition (ii) and continuity,

$$\lim_{m \downarrow 0} \frac{G^{-1}(1-m)}{\frac{1}{m} \int_{1-m}^1 \kappa(u) du} = \frac{G^{-1}(1)}{\kappa(1)} > \frac{G^{-1}(0)}{\kappa(0)} = \lim_{m \downarrow 0} \frac{G^{-1}(m)}{\frac{1}{m} \int_0^m \kappa(u) du}.$$

Hence, there exists $m \in (0, 1/2)$ such that

$$\frac{G^{-1}(m)}{G^{-1}(1-m)} < \frac{\int_0^m \kappa(u) du}{\int_{1-m}^1 \kappa(u) du}.$$

Fix such an m and define the bounded nondecreasing function

$$h(u) := \begin{cases} \frac{1}{2} \left[\frac{G^{-1}(m)}{G^{-1}(1-m)} + \frac{\int_0^m \kappa(z) dz}{\int_{1-m}^1 \kappa(z) dz} \right], & 1-m \leq u \leq 1, \\ 0, & m \leq u < 1-m, \\ -1, & 0 \leq u < m. \end{cases}$$

By construction,

$$\int_0^1 \kappa(u) h(u) du = - \int_0^m \kappa(u) du + h(1) \int_{1-m}^1 \kappa(u) du < 0.$$

By condition (i), we obtain a sequence $\{(\varepsilon_n, F_n)\}_{n=1}^\infty$ such that $\varepsilon_n \downarrow 0$, $F_n \in \mathcal{F}$, and

$$\int_0^1 \left| \frac{F_n^{-1}(u) - Q_0}{\varepsilon_n} - h(u) \right| du \rightarrow 0. \quad (3)$$

Next, for each n , we construct a three-tier mean-preserving contraction $\hat{F}_n^{(3)}$ of F_n by pooling qualities within three quantile intervals. To this end, define

$$Q_{L,n} := \frac{1}{m} \int_0^m F_n^{-1}(u) du, \quad Q_{M,n} := \frac{1}{1-2m} \int_m^{1-m} F_n^{-1}(u) du, \quad Q_{H,n} := \frac{1}{m} \int_{1-m}^1 F_n^{-1}(u) du.$$

Consider

$$\hat{F}_n^{(3)} := m\delta_{Q_{L,n}} + (1 - 2m)\delta_{Q_{M,n}} + m\delta_{Q_{H,n}}.$$

Since $\hat{F}_n^{(3)}$ preserves the conditional means within each quantile interval, $\hat{F}_n^{(3)} \in \text{MPC}(F_n)$ and is feasible by Axiom 1. Moreover, equation (3) implies that

$$Q_{L,n} = Q_0 - \varepsilon_n + o(\varepsilon_n), \quad Q_{M,n} = Q_0 + o(\varepsilon_n), \quad Q_{H,n} = Q_0 + h(1)\varepsilon_n + o(\varepsilon_n).$$

Thus, for all sufficiently large n ,

$$Q_{L,n} < Q_0 < Q_{H,n} \quad \text{and} \quad Q_{L,n} < Q_{M,n} < Q_{H,n}.$$

We now modify $\hat{F}_n^{(3)}$ to obtain a three-tier distribution $F_n^{(3)}$ whose intermediate tier is exactly Q_0 . If $Q_{M,n} = Q_0$, no further pooling is required. We thus consider two cases:

1. If $Q_{M,n} < Q_0$, pool the entire mass at $Q_{M,n}$ with enough of the mass at $Q_{H,n}$ for the pooled mass to have mean Q_0 .
2. If $Q_{M,n} > Q_0$, pool the entire mass at $Q_{M,n}$ with enough of the mass at $Q_{L,n}$ for the pooled mass to have mean Q_0 .

The masses taken from the respective tails are

$$\frac{(1 - 2m) |Q_{M,n} - Q_0|}{Q_{H,n} - Q_0} \quad \text{and} \quad \frac{(1 - 2m) |Q_{M,n} - Q_0|}{Q_0 - Q_{L,n}}.$$

Since $Q_{M,n} = Q_0 + o(\varepsilon_n)$, these masses converge to zero for all sufficiently large n . Consequently, this pooling is possible for all sufficiently large n . We write the resulting distribution as

$$F_n^{(3)} = \pi_{L,n}\delta_{Q_{L,n}} + \pi_{0,n}\delta_{Q_0} + \pi_{H,n}\delta_{Q_{H,n}}, \quad \text{where } \pi_{L,n} + \pi_{0,n} + \pi_{H,n} = 1.$$

Because $F_n^{(3)}$ is a mean-preserving contraction of $\hat{F}_n^{(3)}$, it is feasible by Axiom 1. Moreover, for all sufficiently large n , it has exactly three tiers with strictly positive masses.

Now, consider the mechanism implementing $F_n^{(3)}$. Let $r_{L,n} := G^{-1}(\pi_{L,n})$ and $r_{H,n} := G^{-1}(1 - \pi_{H,n})$ denote the two threshold types. By construction,

$$r_{L,n} \rightarrow G^{-1}(m) \quad \text{and} \quad r_{H,n} \rightarrow G^{-1}(1 - m).$$

The implementing mechanism yields a strict Pareto improvement for all sufficiently large n . By equation (2), the intermediate-tier payment $p_{0,n}$ satisfies

$$-p_{0,n} = \pi_{H,n} r_{H,n} (Q_{H,n} - Q_0) - \pi_{L,n} r_{L,n} (Q_0 - Q_{L,n}).$$

It follows that

$$\frac{-p_{0,n}}{\varepsilon_n} \rightarrow m [G^{-1}(1 - m)h(1) - G^{-1}(m)] > 0.$$

Hence, the intermediate tier is subsidized for all sufficiently large n . By the argument in Section 3.2, $F_n^{(3)}$ therefore yields a three-tier strict Pareto improvement.

B.2 Proof of Lemma 1

In this appendix, we prove that Axiom 3* is weaker than Axiom 3.

Lemma 1. *Suppose Axioms 1 to 3 hold. Then Axiom 3* is satisfied by choosing $\kappa \equiv 1$.*

Suppose that Axioms 1 to 3 hold. Set $\kappa \equiv 1$; this function is continuous and strictly positive. Under this choice of κ , condition (ii) holds:

$$\frac{G^{-1}(1)}{\kappa(1)} = \bar{r} > \underline{r} = \frac{G^{-1}(0)}{\kappa(0)}.$$

To verify condition (i), fix any bounded nondecreasing function $h : [0, 1] \rightarrow \mathbf{R}$ such that

$$\int_0^1 h(u) \, du < 0.$$

By Axiom 3, for every n , there exists $\hat{F}_n \in \mathcal{F}$ such that

$$\mathbf{E}_{\hat{F}_n}[(Q - Q_0)_+] > 0 \quad \text{and} \quad \frac{Q_0 - \mathbf{E}_{\hat{F}_n}[Q]}{\mathbf{E}_{\hat{F}_n}[(Q - Q_0)_+]} \leq \frac{1}{n}.$$

Clearly, $\hat{F}_n \neq \delta_{Q_0}$; thus, Axiom 2 implies that $Q_0 - \mathbf{E}_{\hat{F}_n}[Q] > 0$. Since qualities lie in $[0, 1]$,

$$0 < Q_0 - \mathbf{E}_{\hat{F}_n}[Q] \leq \frac{1}{n} \mathbf{E}_{\hat{F}_n}[(Q - Q_0)_+] \leq \frac{1}{n}.$$

Define

$$\varepsilon_n := \frac{Q_0 - \mathbf{E}_{\hat{F}_n}[Q]}{-\int_0^1 h(u) \, du} > 0. \quad (4)$$

Passing to a subsequence and relabeling if necessary, we may take $\varepsilon_n \downarrow 0$.

Next, observe that $Q_0 \in (0, 1)$. Indeed, if $Q_0 = 1$, then $\mathbf{E}_F[(Q - Q_0)_+] = 0$ for every $F \in \mathcal{F}$, which contradicts Axiom 3. In addition, if $Q_0 = 0$, any distribution satisfying $\mathbf{E}_F[(Q - Q_0)_+] > 0$ would have strictly positive mean—and hence higher average quality than the equity benchmark, which contradicts the definition of Q_0 .

Because h is bounded and nondecreasing, for all sufficiently large n there exists $F_n \in \Delta([0, 1])$ whose quantile function satisfies $F_n^{-1}(u) = Q_0 + \varepsilon_n h(u)$, almost everywhere. By construction, F_n shares the same mean with $\hat{F}_n$:

$$\mathbf{E}_{F_n}[Q] = Q_0 + \varepsilon_n \int_0^1 h(u) \, du = \mathbf{E}_{\hat{F}_n}[Q].$$

We now show that F_n is a mean-preserving contraction of $\hat{F}_n$ for all sufficiently large n . To see this, fix any point t between the lowest and highest support points of F_n . Then

$$\mathbf{E}_{F_n}[(Q - t)_+] \leq 2\varepsilon_n \|h\|_\infty \quad \text{and} \quad \mathbf{E}_{\hat{F}_n}[(Q - t)_+] \geq \mathbf{E}_{\hat{F}_n}[(Q - Q_0)_+] - \varepsilon_n \|h\|_\infty.$$

Moreover, equation (4) implies that

$$\frac{\varepsilon_n}{\mathbf{E}_{\hat{F}_n}[(Q - Q_0)_+]} = \frac{Q_0 - \mathbf{E}_{\hat{F}_n}[Q]}{-\int_0^1 h(u) \, du \cdot \mathbf{E}_{\hat{F}_n}[(Q - Q_0)_+]} \leq \frac{1}{-n \int_0^1 h(u) \, du} \rightarrow 0.$$

Hence, for all sufficiently large n and for all t lying between the lowest and highest support points of F_n ,

$$\mathbf{E}_{F_n}[(Q - t)_+] \leq \mathbf{E}_{\hat{F}_n}[(Q - t)_+].$$

If t lies below the support of F_n , then equality of means gives

$$\mathbf{E}_{F_n}[(Q - t)_+] = \mathbf{E}_{F_n}[Q] - t = \mathbf{E}_{\hat{F}_n}[Q] - t \leq \mathbf{E}_{\hat{F}_n}[(Q - t)_+].$$

If t lies above the support of F_n , then the same inequality holds trivially:

$$\mathbf{E}_{F_n}[(Q - t)_+] = 0 \leq \mathbf{E}_{\hat{F}_n}[(Q - t)_+].$$

Thus, this inequality holds for every $t \in \mathbf{R}$; hence, $F_n \in \text{MPC}(\hat{F}_n)$. Since $\hat{F}_n$ is feasible, Axiom 1 implies that F_n is feasible as well.

Finally, observe that for all sufficiently large n , our definition of F_n satisfies

$$\int_0^1 \left| \frac{F_n^{-1}(u) - Q_0}{\varepsilon_n} - h(u) \right| du = 0.$$

Passing to a subsequence and relabeling if necessary, we obtain the sequence $\{(\varepsilon_n, F_n)\}_{n=1}^\infty$ required by condition (i). Hence, Axiom 3* holds with $\kappa \equiv 1$.